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Thermal Barrier Coatings—A State of the Art Review

Jayant Gopal Thakare[1] · Chandan Pandey[2] · M. M. Mahapatra[3] · R. S. Mulik[1]

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Abstract
Thermal barrier coatings (TBCs) have seen considerable advancement since the initial testing and development of thermal spray coating. Thermal barrier coatings are currently been utilized in various engineering areas which include internal combustion engines, gas turbine blades of jet engines, pyrochemical reprocessing units and many more. The development of new materials, deposition techniques is targeted at improving the life of the underlying substrate. Hence, the performance of the coating plays a vital role in improving the life of substrate. The scope for advancement in thermal barrier coatings is very high and continuous efforts are being made to produce improved and durable coatings. Thermal barrier coatings have the potential to address long term and short-term problems in gas turbine, internal combustion and power generation industry. The study of thermal barrier coating material, performance and life estimation is a critical factor that should be understood to introduce any advancement. The present review gives an overview of the thermal spraying techniques and current advancements in materials, mechanical properties, understanding the high temperature performance, residual stress in the coating, understanding the failure mechanisms and life prediction models for coatings.

Keywords Thermal barrier coating · Residual stresses · Life prediction · High temperature behavior · Spraying techniques

1 Introduction

The operating efficiency of a gas turbine or power generating unit can be improved by retarding the rate and extent of degradation experience by the component. Thermal spray coatings are extensively used for this purpose, a specially designed thermal spray coating called the thermal barrier coating (TBCs) is used to retard the material degradation due to higher heat flux [1]. The application of thermal barrier coating maintains a thermal gradient between the coating surface and the substrate surface thus enhancing the operating temperature range and the efficiency. The development of thermal barrier coatings is therefore vital to enhance the operating temperature range of the power generating unit [2]. The lower the thermal conductivity of the coating material higher will the available temperature range which will increase the overall efficiency of the power generating unit.

Thermal barrier coatings typically consist of a metallic substrate generally a superalloy and followed by an intermediate metallic bond coat used to reduce the thermal expansion gradient between the substrate and the top coat [3]. The top coat is a ceramic coating deposited using plasma spraying technique or electron beam physical vapor deposition (EBPVD). The deposition method effects the properties of the TBCs, the material that are generally used to develop a thermal barrier coating include the NiCoCrAlY, NiCrAlY, NiAl etc. as bond coat material, YSZ, alumina and other advanced ceramics as the top coat material. Post spray processing techniques are also used to enhance the durability and the performance of the TBCs [4].

The objective of this review article is to present an overview regarding the thermal spray coating technology and materials, advancement in thermal barrier coating material and deposition techniques. The articles also present the properties of TBC system, residual stress in thermal barrier

✉ Chandan Pandey
jscpandey@iitj.ac.in; chandanpy.1989@gmail.com

Jayant Gopal Thakare
jayantgt88@gmail.com

M. M. Mahapatra
manasmohan2@gmail.com; mmmahapatra@iitbbs.ac.in

[1] Department of Mechanical and Industrial Engineering, IIT Roorkee, Roorkee 247667, India

[2] Department of Mechanical Engineering, Indian Institute of Technology Jodhpur, Jodhpur, Rajasthan 342037, India

[3] School of Mechanical Sciences, IIT Bhubaneswar, Bhubaneswar, Odisha 752050, India

coating, high temperature performance of thermal barrier coating and advancement in life prediction of thermal barrier coating.

2 Application of Thermal Spray Coatings

The structural components of a power generating unit mostly operate in a hostile and corrosive environment which limits the component life [5]. The high temperature, pressure and the presence of flue gases accelerates the degradation of the metal components. This degradation may result in heavy losses resulting from the reduced efficiency and maintenance. The degradation can also lead to premature failure of the components. The failure of components can be avoided by the use of protective coatings or claddings which are resistant to high-temperature degradation [6]. However, the use of cladding is not recommended as it changes the microstructure of the underlying substrate which makes it susceptible to other modes of failure [7]. The use of a coating, on the other hand, turns out to be the safest mode of protecting the components against high-temperature degradation and enhancing the life of the component.

The surface degradation of high-temperature components is a lifetime determining factor for a unit operating at high temperatures. The use of the coating increases the lifetime of the substrate by protecting the surface from the aggressive oxidizing environment. This has resulted in appreciable increase in use of coatings in power generation units, oil and gas refineries, waste incinerators plants and other recycling, reprocessing units [8]. The use of coating has reduced the down time resulting from plant break down, increasing the overall efficiency of the unit. The coatings are deposited by various techniques which have been broadly classifiedinto gaseous state processes, solution state processes, molten or semi-molten state processes, and solid-state processes based on the physical states via which the coating is deposited. The gaseous state processes include physical vapor deposition (PVD), chemical vapor deposition (CVD) and ion beam assisted deposition (IBAD) process [9–13]. The chemical solution process, an electrochemical deposition process, and the sol gel process are the type of solution state processes. Welding, thermal spraying and laser technique are the processes that deposit the coating in melted and semi-melted condition. The most versatile process which can be applied to components of any dimension is the thermal spray coating process. This coating process does not alter the microstructure of the substrate, unlike welding and laser cladding process.

3 Thermal Spray Coating

3.1 Introduction

Thermal spraying is a process in which un-melted, partially melted, and completely melted feedstock particles are deposited on a pretreated substrate surface [14]. The process involves generating a stream of feedstock particles which deform plastically on bombarding the pretreated substrate surface as shown in Fig. 1. The surface is pretreated to produce adequate surface roughness in order to anchor the incoming feedstock powder particles [15]. The coating deposited on the substrate surface with aim to enhance the wear resistance, minimize surface degradation via corrosion, thermal cycling etc. The coating material is selected such that the coefficient of thermal expansion of the two materials is mutually compatible [16]. An intermediate metallic bond coat is preferred when the CTE difference between the substrate and the coating is large.

3.2 Thermal Spraying Techniques

The thermal spraying techniques are broadly classified based on the mode of energy input to the carrier gasses. The Fig. 2 shows the broad classification of thermal spraying processes based on the energy input to the gases [17]. Chemical energy released via combustion and electric energy released by formation of arc are the two main sources of energy that elevate the thermal energy of the particles.

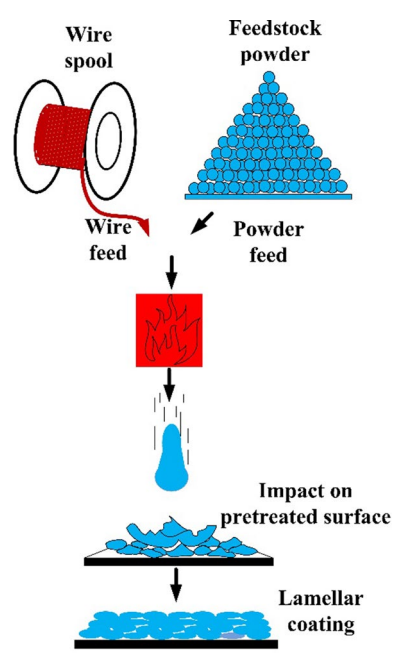

Fig. 1 Schematic for thermal spray coating process

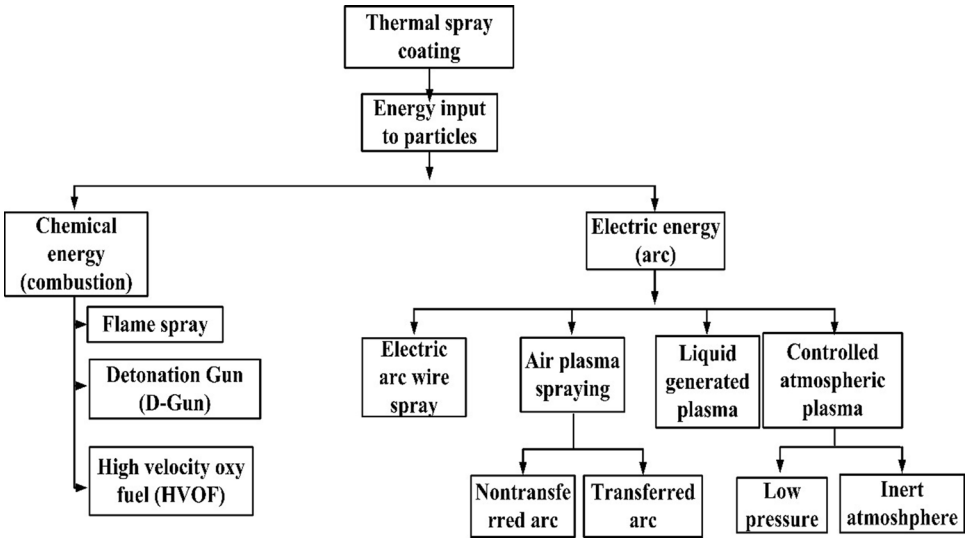

Fig. 2 Broad classification of thermal spray coating techniques based on input energy

An innovative advancement in the coating process called the cold spray technique does not involve providing thermal energy, the whole process of coating is executed under the influence of kinetic energy [18–23].

The Fig. 3 illustrates the classification of the thermal spraying process, every thermal spray coating process is unique in terms of the energy it provides to the powder particles and hence coating developed by difference processes has unique properties. The thermal spray coating techniques that are commonly used are explained in brief as below.

Flame spraying involves use of thermal energy generated due to the combustion of a fuel gas generating a flame as shown in Fig. 4. The fuel gas inlet is axial and the feedstock in form of powder or wires or rods are fed either axially or perpendicular to the flame. The feedstock melts to form molten droplets which are atomized by compressed air and carried to the substrate surface with appreciable kinetic energy. This coating process is used for depositing metals, alloys, and polymers. Spherical morphology of powders is preferred to facilitate the handling of powder. The air to fuel ratio and total gas flow are few parameters that are adjusted to control the thermal profile of the flame. The jet speed is of the order of 100 m/s and the particle velocity can rise to 80 m/s prior to impact. The coating deposited using flame spray technique has high porosity content of around 10%–20% [24, 25].

Detonation gun (D-Gun) and high velocity oxy-fuel (HVOF) technique are other two variants that employ the chemical energy released as a result of the fuel combustion. These two variants differ in the way the fuel is burned, continuous burning is carried out in HVOF technique whereas the burning is intermittent in D-gun process. In D-gun process shown in Fig. 5a, an explosive mixture of fuel, feedstock powder, and oxygen are fed into a narrow barrel [26,

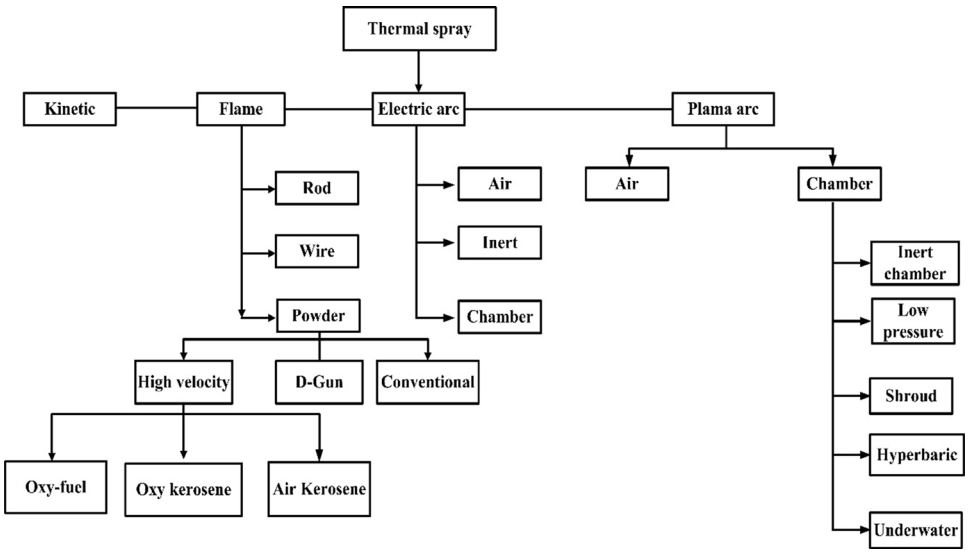

Fig. 3 Detailed classification of thermal spray coating processes

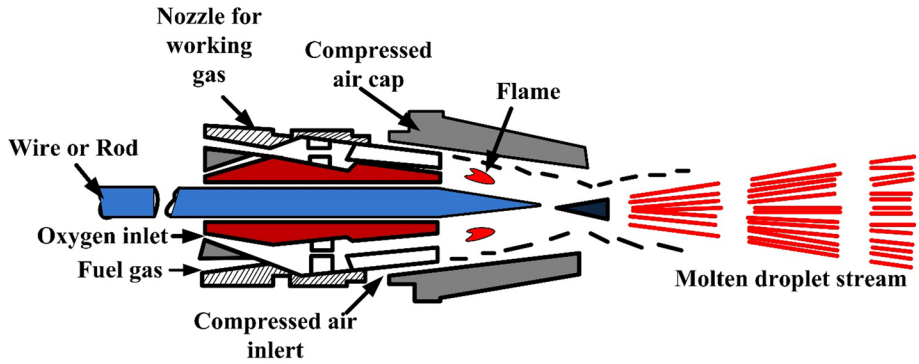

Fig. 4 Schematic of flame spray technique

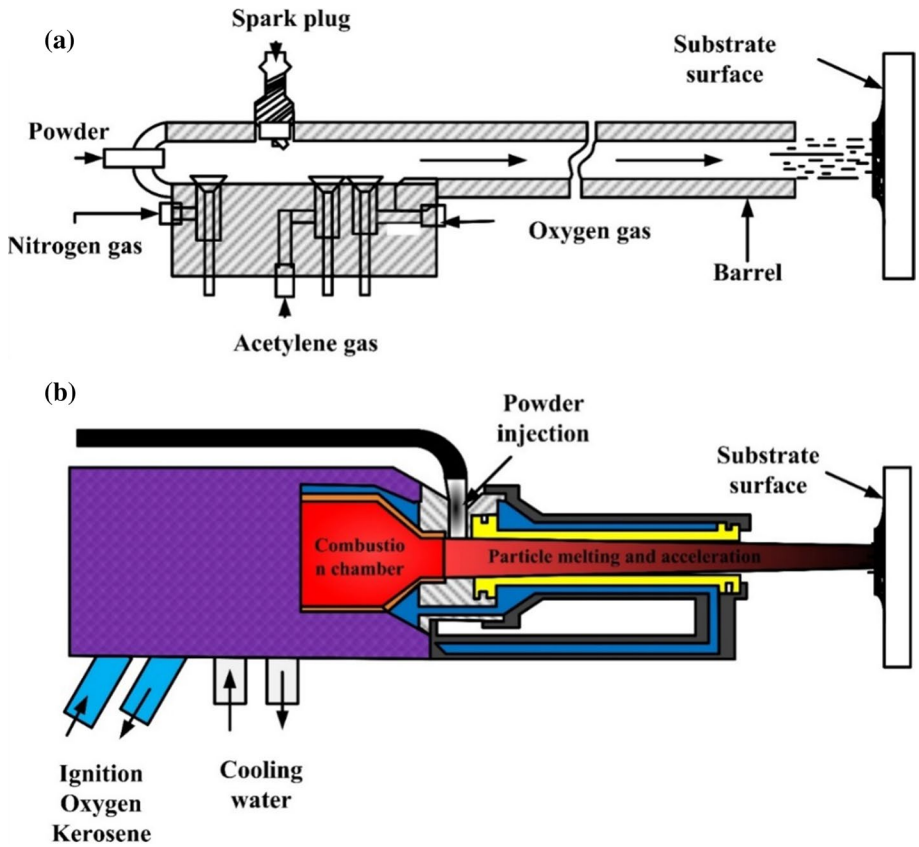

Fig. 5 Schematic showing **a** D-Gun process, **b** high velocity oxy fuel technique

27]. The narrow barrel constraints the explosive mixture which is detonated using a spark plug resulting in formation of pressurized heat wave. This heat wave travels down the water-cooled barrel carrying the powder and depositing on the pre-treated surface. Nitrogen is used to purge the barrel in between the detonations which occur at a frequency of 3 to 6 Hz. The detonation gun produces a highly pressurized wave by which the particles attain a velocity of the order of 800 m/s. These high particle velocities on contacting the surface form a coating with high bond strength and better density [28]. In HVOF process the combustion is continuous, the fuel and oxygen are introduced in the water- or air-cooled combustion chamber. The exhaust and the flame make an exit from the nozzle with a supersonic velocity, the powder is introduced radially or axially in the flame as shown in Fig. 5b.

Air plasma spray (APS) process uses plasma to impart thermal energy to the particles. A plasma is hot ionized gas produced by transferring energy into the gas until the electrons, positive ions are formed [29–31]. The electrons and the ions recombine once the energy input is removed releasing large amount of thermal energy. When a certain amount of energy is supplied to the molecules of a gas, the molecules dissociate into atoms due to mutual collision. When very high magnitude of energy is supplied to a gas, the electrons from the atomic shell are also forced out thus ionizing

the gas. The electron will have the tendency to return to its original lower energy state releasing energy in the form electromagnetic waves.

The plasma generator is shown in Fig. 6, consists of an anode and a cathode the electric arc is generated in the constricted space. The anode is usually made of copper and a thoriated tungsten is used as cathode, the arc heats up the working gas which expand to form a jet. The carrier gas feeds the power into the jet which melts the powder and accelerates the particles towards the pretreated substrate surface. Argon gas is the most commonly used working gas, however a mixture of Ar, and hydrogen, or helium, or nitrogen is also used. The voltage is between 30 and 70 V and the temperature and the velocity attained at the exit of the nozzle is around 14,000 and 800 m/s. The velocities of the order 2500 m/s have been attained by using a high-power plasma spray installation (Plazjet). There are different variations of plasma spray process depending on the environment in which the spraying is carried out like the low-pressure plasma spraying, or vacuum plasma spraying, controlled atmospheric plasma spraying (CAPS). Plasma consists of partially ionized gas with high kinetic energy and high temperature [32]. In a vacuum plasma spraying process, the coating process is carried out in low pressure environment [33]. In CAPS the plasma torch is similar to the APS torch and the spray chamber can withstand a pressure of 0.4 MPa. The few other variants of plasma spray process are inert plasma spraying (IPS), shrouded plasma spraying (SPS), atmospheric temperature-controlled spraying (ATCS) [34, 35].

The ceramic coatings are found to be highly resistant to high temperature degradation and hence are commonly used as thermal and environment barrier coating (T/EBC) [36]. The requirement for high gas turbine inlet temperature has popularized the use of thermal barrier coatings. The T/EBC is also predominantly used in internal combustion engines, turbine blades, vanes, and other hot sections of a power generating unit. The architecture of a T/EBC consists of a substrate, a metallic bond coat and a ceramic top coat [8]. The most commonly used bond coat is nickel, cobalt or iron based MCrAlY metallic coating where "M" represents the main constituent of the bond coat. The aluminum in the coatings facilitates the formation of aluminum oxide which acts as a diffusion barrier to the detrimental species. The top coat is an air plasma sprayed ceramic coating usually a yttria (7%–8%) stabilized zirconia (YSZ) coating. The air plasma spray (APS) technique inherently produces porous ceramic coatings, the interconnected pores act as a pathway for detrimental species. The use of low-quality fuel (containing Na & V) for combustion results in formation of low melting salts like Na_2SO_4 and V_2O_5 on the coatings [37–39]. The molten saltreacts aggressively with the coating and the underlying substrate. The YSZ coating deteriorates as a result of the selective leaching of yttrium leading to degradation of the coating. The drawback of an APS YSZ coating is the lack of resistance to the infiltration of detrimental species.

3.3 Thermal Spray Coating Materials

Thermal spray coating materials can be classified based on the architecture of the coating and the coating material. Three basic materials can be successfully thermally sprayed namely single-phase materials, cermet and composites. The single-phase material comprises of metals, alloys, Intermetallics, polymers and ceramics, whereas the composite materials are cermet, reinforced ceramics, reinforced polymers, reinforced metals. The coatings are also classified based on the architecture of the coating as single layer, bi-layer and functionally graded coating [8]. The metal and alloy coatings mainly include material which are used to repair the base metal and to improve the surface properties of the base metal. The most commonly sprayed

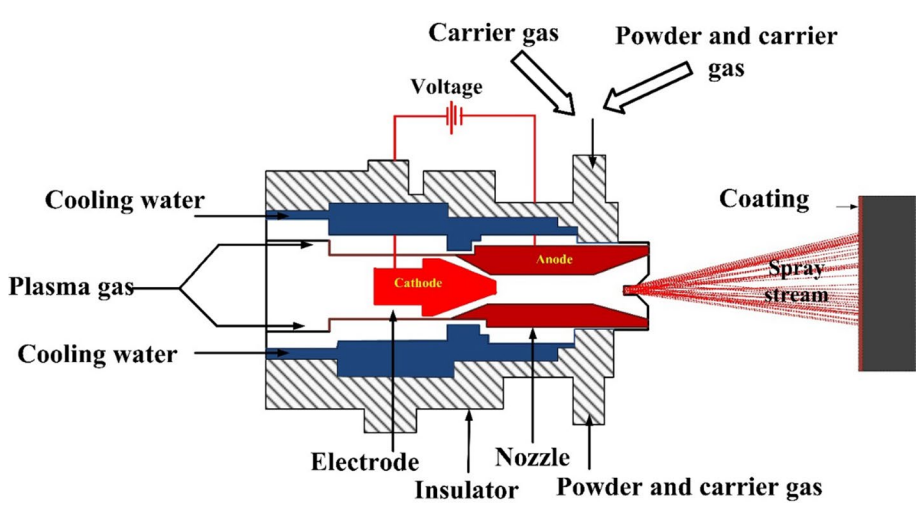

Fig. 6 Schematic representing plasma generator and plamsa spray coating process

metals and alloys include, stainless steel, cobalt, nickel, aluminum, bronze, zinc, molybdenum, NiCrAlY, NiCoCrAlY. Metal and alloy coatings are mainly used in piston rings, engine cylinders, bridges, petrochemical equipment and other corrosion prone applications. The ceramics are also thermally sprayed and are popular due to their high thermal insulation, electrical insulation and oxidation resistance. The most commonly used ceramic coating include Al_2O_3, ZrO_2, YSZ, Cr_3C_2, and TiC etc. The ceramics generally have a very high melting point hence are most suitable to be sprayed by plasma spray techniques. Intermetallic are highly sensitive to oxidation at elevated temperature and hence should be sprayed in controlled or inert atmosphere [17]. Intermetallics are metallic alloys that exist in the form of compounds having an ordered crystal structure, high melting point and a brittle behavior.

Polymers are thermally sprayed using conventional flame spray and HVOF technique. Thermosetting plastics, thermoplastics, and copolymers like polytetrafluoroethylene (PTFE), poly-methyl-methacrylate (PMMA), polyimide etc. are commonly sprayed polymers. Composite materials are classified as particulate reinforced, fiber reinforced, and whisker reinforced composite. The cermet are ceramic particles dispersed in a metallic matrix and most common cermet coatings are WC/Co, Cr_3C_2/NiCr, and the metal matrix acts as binder for the ceramic particles.

4 Thermal Barrier Coating System

4.1 Introduction

A thermal barrier coating typically consists of a high temperature structural material base, an intermediate metallic bond coat and a ceramic top coat. The ceramic top coat acts as an insulator and resists the heat transfer and also provides protection against high temperature degradation of the underlying substrate. The ceramic top coat is commonly deposited using electron beam physical vapor deposition (EB–PVD) or APS or any modified version of any of the two. In air plasma spray process the heating of substrate is not mandatory, however heating may be carried out to reduce the residual stress in the coating [40–44]. In case of EBPVD the preheating of the substrate is compulsory, and the rate of coating deposition is very low as compared to that of the plasma spraying process. The coatings produced by the two-process exhibit variation in the microstructure. The coating deposited via APS process exhibit a lamellar morphology whereas the coating deposited using EBPVD process exhibits a columnar morphology.

EBPVD TBCs can be classified into; two layered and functionally graded coating. A two layered coating consists of a MCrAlY bond coat (M=Ni, Co) and YSZ (6%–8%) as top coating [45]. A functionally graded TBC consists of bond coat and multiple layers of composite powder which tend to reduce the coefficient of thermal expansion gradient between the top coat and the bond coat [46]. The functionally graded thermal barrier coatings are used when the difference in CTE between the coating constituents drastically affect the performance of the coating.

In case of APS deposited coating porosity and defects in the form of minor cracks is an inherent part of the coating [47]. The presence of these defects reduces the thermal conductivity of the already low conducting YSZ. The APS techniques are predominantly applied as compared to the coating deposited using EB–PVD. The APS sprayed coating has another advantage in terms of the coating thickness, thick coating can easily be produced by APS coating. The coating microstructure of an APS sprayed coating consists of a complex porous lamellar structure [32].

The basic unit of a lamellar APS coating is a pancake shaped splat of thickness depending on the spray parameters can vary from 1 to 50 µm with diameter of the order of 10–50 µm [48]. This splats extent both along the plane and out of the plane of the substrate, the discrete splats overlap to form a coating, however a distinct boundary exists between the splats as shown in the schematic Fig. 7. The plasma sprayed coating shows the presence of pores with variable

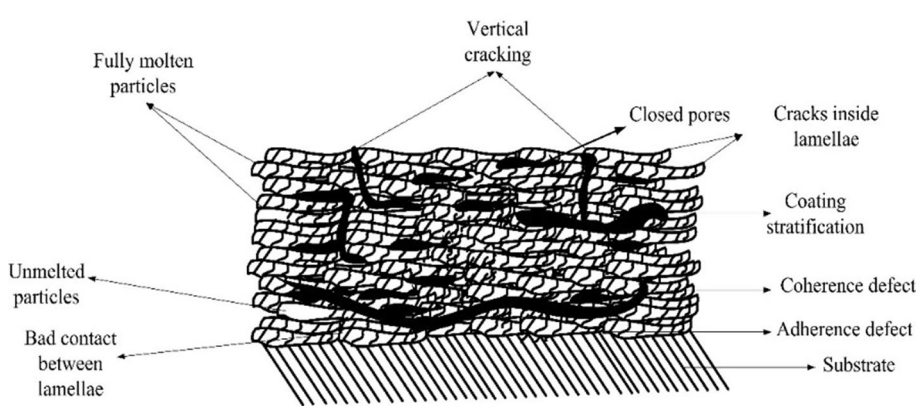

Fig. 7 Schematic of a typical plasma sprayed coating exhibiting lamellar structure

dimensions, cracks are also present in the form of horizontal and vertical micro defects [49]. The delaminating crack can be intra-lamellar or inter-lamellar. The vertical microcracks are cause as a result of lack of adhesion between the lamellas and the splats. Some of the cracking is also caused due to quenching of the coating constituents, resulting in relaxation of residual stress [50]. Many properties like adhesive strength of the coating are affected by the splat structure. Splat flattening is dependent on the impact velocity and heat input to the feedstock particles and is affected by the surface wettability, roughness and reactivity of the surface. The porosity in an APS coating increases if the wettability of the coating is lower [51]. Higher surface roughness of the substrate results in development of thick coating. The substrate preheating also affects the impinging splat, splat quenching results in restricted contraction resulting in formation of tensile residual stress [52]. Functionally graded coating might be helpful in reducing the residual stress in the coating. The functionally graded coating refers to the layered structure of the coating, which result in gradual change in thermal expansioncoefficient between the constituents of the coating and the substrate as in the case of a functionally graded coating [12].

The thermal barrier coatings ((TBCs) are typically consists of a bond coat and a top coat. MCrAlY (M: nickel, cobalt, or a mixture of both elements) and heat resistant material, such as yttria stabilised zirconia (8YSZ) are the most commonly used material for the bond coat and top coat, respectively. To deposit the bond coat and top coat on metallic substrate, a thermal spray techniques are employed [34, 53].

A large number of porosities has been noticed for the plasma sprayed 8YSZ coating which allow the diffusion of the oxidizing species that leads the oxidation of the bond coat and resulted in the formation of thick thermally grown oxide (TGO) layer. The excessive growth of TGO layer leads the de-cohesion in the bond coat and resulted in final fracture of the TBC [54, 55]. A number of research works have been published to evaluate the performance of the coating in the harsh working environment. The performance of the coating is mainly evaluated in terms of the degradation in microstructure [31, 56–58]. Hence, by altering the microstructure of the coating in bond coat might be helpful in improving the durability. The altering of the microstructure mainly deals with the controlling of the porosity and it can be achieved by varying the coating deposition process and post thermal spray treatment of the coating [59].

The current work mainly deals with the development of the coating by nano-size ceramics powders, cermet powders, and powders doped with rare earth oxides and these modification are mainly performed to alter the microstructure of the coating [54, 60–63]. Addition of the Alumina in YSZ powder is done to produce a coating with increased resistance to diffusion of detrimental species. Resistance to the diffusion resulted in the reduction in thickness of thermally grown oxide and finally reduces the chance of the failure of coating due to excessive growth of thermally grown oxide [59, 64, 65].

4.2 Properties of Thermal Barrier Coating

There are certain properties that must be analyzed and considered to study the performance and durability of a thermal barrier coating. The properties include the elastic–plastic properties, cohesive strength, adhesive strength, fracture properties, thermo-mechanical fatigue properties, wear, and erosion. This properties of thermal barrier coatings vary with change in torch design, feedstock characteristics, and operating parameters [66]. The microstructure and the properties of the TBC is predominantly dependent on the process parameters. The properties of the coating are also affected by the interaction between the bond coat, top coat and the substrate at higher temperature. The TBC microstructure is non homogeneous and the properties are likely to be affected by the measurement techniques. The properties can be measured conventionally via bend test, tensile test, adhesion test [67–72]. Indirect measurement methods like nano-indentation, micro indentation are also effectively used to determine the properties of TBCs [73]. The properties can be measured on free standing coating or coated specimens, the properties exhibit variation for free standing coating and coated specimen.

The TBCs are subjected to intensive thermal cycling, thermal fatigue is a major cause of coating failure. The thermal stress cause cracking in the coating which eventually leads to delamination of the ceramic top coat. The elastic modulus of the top coat is fundamental in determining the resistance of the coating to crack propagation. The elastic modulus of the top coat is around10 GPa, 100 GPa and 200 GPa when measured via bending, micro indentation and nano indentation respectively [24, 26, 32]. The coating properties are affected by thermal exposure as a result of high temperature sintering. The sintering results in closure of micro-defects like cracks, and voids resulting from the grain growth across the splats.

The time–temperature dependent plastic properties of the bond coat should also be analyzed to understand the failure of the TBCs. The tensile stress in the bond coat are relaxed during the heating part of the thermal cycling, the oxide growth also introduces stress during the exposure. The plastic properties of the bond coat are vital in preventing the fracture in the brittle interface. The creep strength of the bond coat should be sufficiently high to avoid the negative effects of compressive stresses induced in the TGO [75]. An in-plane compressive stress is setup in the top coat when the cooling rate is very high, this

compressive stress induces an out of the plane tensile stress which causes spallation of the top coat. A less ductile and weak bond coat causes instability and buckling of TGO which is followed by TGO cracking [76]. A durable bond coat material should exhibit intermediate ductility and strength at both elevated temperatures and low temperatures. MCrAlY coatings have higher ductility as compared to aluminide coatings and hence are considered ideal for bond coat applications.

The operation and duty cycles in actual conditions is simulated in laboratory using burner rigs tests. The thermal shocks and the accelerated thermal cyclic tests correspond to the actual start up and shut down conditions. This accelerated thermal cyclic tests lead to spallation of the bond coat and top coat without affecting the substrate. Fracture properties of both the bond coat and the top coat are very critical to with stand this severe thermal fatigue conditions. During low cycle fatigue testing conditions cracks have been reported to be generated in the TGO or in the metallic bond coat [77]. The crack is directed perpendicular to the plane of bond coat moving towards the substrate. The crack generated in the TBC system has to propagate through the bond coat to render the TBC system useless. The fatigue properties are hence very critical in understanding the performance of the TBC system.

There are several parameters beyond those discussed above also affects the selection of the coating material [11]. The basic requirement of the coating material is to offer the high resistance to the reaction with the oxidizing specimen. The difference in thermal expansion coefficient between the coatings and substrate material should be marginal to stop any type of the delamination or cracking comes in the picture due to the CTE mismatch. The other factor is the stability of the coating material at elevated service temperature condition and the coating constituents should also be chemically inert to the substrate.

To overcome the gradient in CTE between the substrate and top coat, TBCs and EBCs are comprises of two-layers, first is metallic bond coat and followed by the ceramic top coat. The deposition of the ceramic top coat is performed after the deposition of the metallic bond coat [78]. Plasma spray technique are most commonly employed for the ceramic coatings [79]. However, plasma sprayed coatings inherently consist of the high porosities which allow penetration of oxidizing species and the excessive infiltration of the species leads to oxidation of bond coat and formation of thermally grown oxide (TGO) layer. The growth of the TGO layer thickness beyond a certain point has been recognized as the main reason for the failure of TBCs [54, 55]. Thedegradation in coating properties in the aggressive environment occurs mainly due to microstructure instability [31, 54, 56]. The coating performance in elevated service condition is mainly improved by altering the porosity, deposition parameters, feedstock composition, and post thermal spray treatment [59, 80].

In current years, an accelerated progress is reported in research and development of ceramic matrix composites, TBCs, EBCs, and solid oxide fuel cells [81]. The performance of the coating is improved by altering the altering the microstructure, spray technique, architecture of the TBCs. As compared to conventional YSZ coating, a high performance 4 YSZ coating has been produced that offers improved mechanical behavior and thermal properties as compared to conventional YSZ [81, 82]. A two differed triple layered architecture is developed by Mahade et al. [83] to improve the TBC performance against CMAS (calcium magnesium alumina silicate) attack. Richards and Wadley [84] has been also produced a tri-layer Yb2SiO5 coating on a Si substrate using air plasma spray coating (APS). A double layered coating consists of the ZrB_2, $MoSi_2$, and SiC were developed by the APS technique and an excellentablation resistance in high-frequency plasma were reported [85]. The effect of the doping of the some rare earth oxides, metals such as CeO_2, ln_2O_3, and Sc_2O_3 on the performance of YSZ has also been evaluated and improved performance is reported after doping [86, 87]. The coating for advanced materials like La_2ZrO_7 are relatively less susceptible to molten salt at 1000 °C [88]. An improved mechanical properties, improved corrosion resistance and oxidation resistance have also been reported for the coating produced by the nanostructure feedstock [82, 89–91]. Yu et al. [92] had produced the boron alumina-silicate glass–ceramic coatings on the titanium substratefor varying concentration of the MoO_3 and improvement was reported around 96% for 100 h oxidation at 850 °C. The doping of the alumina in YSZ coating is performed to enhance theresistance to diffusion of the detrimental species that reduces the TGO layer thickness and increases the life of the coating [30, 64].

4.3 Residual Stress in Thermal Barrier Coating

The application of coating in industrial applications is governed by the quality of the coating. The residual stress generated in the coating during the deposition phase can result in premature failure of the coating by spallation and cracking [93, 94]. The initial stress state in the coating must be evaluated to predict the behavior of the coating under cyclic thermal or isothermal loading conditions [95, 96]. Destructive and non-destructive techniques have been developed to measure the residual stress in metallic and ceramic coatings. The residual stress generated in the thermally sprayed coating is a factor that invariably affects the life of the coating. The combined effect of the residual stress and the service stress results in failure of TBCs. The presence of residual stress in thick thermal barrier coating leads to generation of segmentation cracks which are visible along the cross

section immediately after the deposition of the coating [97, 98].

Residual stress in thermally sprayed coating is generated during the deposition process when hot particles (melted or semi-melted) come in contact with a relatively cooler substrate or a pre-existing coating lamella. The residual stress in the coating is generated as a result of two main sources, quenching of deposited particles and difference in coefficient of thermal expansion (CTE) between the deposited coating and the substrate. The quenching stresses are generated in the coating when the particles in flight come in contact with the substrate and experiences sudden cooling which is referred as primary cooling. The residual stress is also introduced when the coating substrate system cools from deposition temperature to room temperature referred as secondary cooling [99]. The quenching stresses introduced due to primary cooling are tensile in nature and the residual stresses due to CTE mismatch are compressive in nature when the coating CTE is lower than that of the substrate and if not, the stresses are tensile in nature. The magnitude of residual stresses in the coating is also affected by inflight velocity of particles and particle size. The growth of thermally grown oxide and the influence of hot corrosion also effects the level of residual stress in the coating [100]. The presence of compressive residual stress are favorable whereas the presence of tensile stresses leads to easy initiation and propagation resulting in delamination and loss of adhesion [101]. The estimation of residual stress is highly significant in evaluating the life of coating and the structural component. Residual stress is estimated by various techniques which can be classified under two main categories namely non-destructive and destructive. The X-ray diffraction, neutron diffraction are few of the nondestructive techniques and whereas the incremental hole drilling method, curvature method, modified layer removal method and other experimental techniques are few examples of destructive techniques [102–105].

The non-destructive methods are based on the measurement in variation of the crystal lattice spacing (d) as compared to the unstressed crystal lattice. The diffraction peaks exhibit a peak shift when measured at different orientations with respect to the specimen surface. The stress is determined from the strain measurements using elastic constants. Residual stress determination using indentation is also a potential nondestructive testing technique. The load–displacement curve is continuously monitored to determine the residual stress in the coating.

The destructive testing techniques rely on the strain released as a result of the material removal. In the hole drilling (HD) and layer removal method (LR) released strain is recorded after every subsequent cut using strain gauge attached to the coating or the substrate. The layer removal method was modified by Greying et al. [106] to evaluate the through thickness distribution of residual stress. The curvature method is one of the experimental techniques used to analyze the average stress value of the thermal spray coating. The residual stress was measured to be 200 MPa in Ni–Al_2O_3 composite coating using curvature measurement method [107]. The curvature method is very often used to determine the residual stress in the coating. However, the curvature cannot be measured very accurately. Material removal using chemical dissolution is have also been used to determine the residual stress in 8YSZ thermal barrier coating [108]. The residual stress measured in WC–10%NiCrBSi coating using hole drilling was found to be compressive. The magnitude was found to increase towards the interface [103]. A precision hole drilling machine is used to control the depth during drilling. The compressive stress at the interface is reported to be higher than that at the free surface of the top coat. The radial stress is reported to be around 190 MPa in the substrate and 150 MPa in the ceramic coating. Advanced micro drilling and milling process is also used to determine the residual stress in the coating. The combination of drilling and milling operating at high drilling speeds, low feed rate and low drilling depth ensures negligible amount of induced stress in the during material removal process. The coating was found to have compressive residual stress which increased with the increase in preheating temperature. The coating also showed presence of tensile stresses, however the stresses were between 10 to 30 MPa [109]. The residual stress in high velocity oxy fuel sprayed nickel based coating determined using incremental hole drilling method was found to be tensile in nature with the magnitude between 150 and 300 MPa [105].

The fracture properties of thermal barrier coating are dependent on the state of residual stress in the coating. The presence of lower compressive stress reduces the critical strain required to generate the first transverse crack. The substrate preheating temperature also affects the residual stress in the coating. A higher preheating temperature and deposition temperature, coating deposited at high temperature result in formation of higher compressive residual stress [110]. The increase in compressive residual stress results in increase in fracture toughness of the coating. The deposition temperature not only affects the residual stress at coating surface but also affects the elastic modulus of the coating. The elastic modulus monotonically increases with increase in the deposition temperature. The elastic modulus in the coating increases due to two main reasons firstly the increase in deposition temperature increase the adhesion between the molten splats. The higher compressive stress resulting from the higher deposition temperature suppresses the micro crack initiation and inter splat sliding. The proportion of quenching and thermal stresses governs the final state of stress in the coating. The average value of strength is reported to increase from 11.2 MPa to 73.6 GPa with an increase in deposition temperature from 186 to 748 °C,

the fracture toughness of YSZ was reported to vary from 5 to 80 MPa [74, 111]. The fracture strength also increases with increase in the ration of residual stress and the elastic modulus of the coating. The effect of residual stress can be evaluated by the effect on bending strength of the coating. The coating with high compressive residual stress when subjected to external tensile stress less stress concentration will occur at the voids and other defect sites leading to lower cracking tendency.

4.4 High Temperature Behavior of Thermal Barrier Coating

The performance of a power generation plant is defined by the efficiency of the unit and the energy saving it provides to the unit. Increasing the turbine inlet temperature increases the efficiency of the power generating unit, however the materials of the structural components have specific operating limits [112]. The increase in temperature should be accompanied with an improvement in material properties. The components operating at high temperature are subjected to variety of loads which include thermal and mechanical loading. This thermo mechanical loading is accompanied with the presence of chlorides, sulfides, slats and erosive particles, making the operating environment highly detrimental to the structural components. The operating conditions also involve deposition of molten salt on the surface of the coating which finally leads to delamination and failure of the coating [113]. Thermally sprayed ceramic coating exhibit high thermal insulation, oxidation resistance, and enhances the life and performance of the structure [114, 115]. The thermal barrier coating typically consists of a metallic bond coat and a ceramic top coat. The bond coat is predominantly metallic and Ni, Co, Cr, Al, Ta, Si, Hf, Y, Y_2O_3, Pt are the most common constituents of the bond coat [116]. The factors that affect the oxidation resistance of the bond coat are, the interdiffusion between the bond coat and the substrate, microstructure and the morphology of thermally grown oxides (TGO), cracking resulting from the phase transformation [117–119]. In a TBC the most commonly used bond coats are MCrAlY (M=Ni/Co, NiCo) and platinum doped aluminides. A metal undergoes oxidation when either the metal atoms experience an outward diffusion or oxidizing species diffuse inward and forms a metal oxide interface. At higher operating temperatures metals forms subsurface precipitates instead of a discrete oxide layer [116]. The stresses are expected to generate due to the oxide formation as the oxide volume is different from the consumed metal [120].

The performance of power generating units at high temperature service exposure condition is strongly influenced by the hot corrosion (HC) resistance of metals [121]. In high-temperature operating power units, sections like gas turbines, gas combustion chambers, inlet pipes, steam boiler tubes etc. are subjected to HC [122–124]. To meet the advancement of the materials and high temperature service condition, creep strength enhanced ferritic-martensitic steels are developed [125]. At high temperature service condition, the degradation in life of that steel components are commonly reported in literatures and it is governed by the oxidation and hot corrosion [124, 126]. The improved service life at high temperature can be achieved by increasing the HC behavior of materials.

To resist the degradation in service life or mechanical properties, these components are subjected to metallic, cermet and ceramic coatings on the surface [39, 127, 128]. Ceramics are commonly employed for thermal and environmental barrier coatings (T/EBC) to resist the degradation and enhance the operating temperature of gas turbines [129–131]. 6–8 wt% Yttria-stabilized zirconia (6–8% YSZ) are most commonly employed ceramic coating and the deposition on a metallic bond coat is carried out by air plasma spray technique (APS) or electron beam physical vapor deposition (EB–PVD) [44, 132].

Initially stage of the oxidation leads the formation of thermally grown aluminum oxide and after the complete depletion of the aluminum from the bond coat, Ni/Co based oxides and spinel's formation occurs. The major failure occurred due to the excessive growth of TGO and spinel's formation. A premature cracking of the coating was also reported and it occurs mainly due to the severely corrosive working environment [133]. The premature failure of the YSZ based TBC system was reported in presence of the impurities like Na, V, S that leads the formation of low quality combustion product like Na_2SO_4, V_2O_5 [134]. That impurities react the ceramic top coat and reduces the durability of the TBC material [135, 136]. A very aggressive reaction is reported between V2O5 and zirconia-based TBC that results in the premature failure of the coating [20]. The air plasma sprayed conventional YSZ coating is characterized by a porosity range of 10%–25% [137].

In recent years, research is focused on improving the service life of the coating and HCbehavior of the conventional YSZ based TBC and it is achieved by altering the deposition process, microstructure, architecture of the coating system and feedstock composition [138].

The HC behavior of the GdYb-YSZ coating was performed by the Guo et al. [138]. Guo et al. [138] observed that the presence of Yb_2O_3 acted as phase stabilizer and retarded the phase transformation of zirconia. The presence of CeO_2 in LZ_7C_3 coating makes it more resistance to HC attack as compared to $La_2Zr_2O_7$ and conventional YSZ coating [138].

The functional behavior of the coating is strongly influenced by the microstructural behavior and a better life is reported for the columnar microstructure as compared to

the lamellar morphology. Electron beam physical vapor deposition promotes the formation of columnar morphology while lamellar morphology formation was obtained for the thermal spraying process. Suspension plasma spray technique might also be helpful to obtain a columnar microstructure using sub-micron sized powders [29]. Attempts have also been made to improve the HC behavior of the conventional YSZ coating by engineering the architecture of the coating.

The improved HC behavior is also reported for the multilayered composite coating with varying percentage of alumina exhibits as compared to the conventional YSZ coating [139]. Gadolinium zirconate, used as thermal barrier coating with YSZ base layer showed superior thermal cycling behavior and HC behavior as compared to YSZ based TBC and single layered GZ coating [8]. The post-weld treatment like laser glazing, sintering may also improve the HC behavior of coating for APS coating [8]. The post-processing does not ensure improvement in the performance of the coating because it does not ensure complete closure of cracks.

The delamination failure of the coating exposes the underlying bare metal surface which ultimately leads to premature failure of the structure. The performance of the coating in harsh environment with and without the presence of molten salt should be analyzed to avoid any premature failure of the component. The understanding of degradation mechanism is very important to prolong the life of the coating and the structural component. The hot components operate in presence of oxygen containing flue gases which convert the metallic coating in metallic oxides which are referred as thermally grown oxides (TGO) [140]. The TGOs act as a diffusion barrier and retard the infiltration of the oxidizing species. In alumina based metallic bond coat Al_2O_3 is the major component of the TGO and provides protection up to the melting point of underlying substrate whereas Cr_2O_3 are protective up to 871 °C [140]. The TGO can be formed and maintained above a critical level when the Al, Cr, or Si does not fall below a critical level of around 4–5 at% [123]. The presence of sulphur in the operating environment reduces the life of the TGO and the incorporation of rare earth oxides in the bond coat improves the adherence of the coating. The metallic oxides should satisfy few criterions to render it protective [116]. The oxide should be thermodynamically stable and should form a adherent, uniform thick layer at a slow growth rate. The stability of the oxide is related to its ability of resist any undesirable phase change, lower rate of diffusion across the interface, minimum brittle phase formation and compositional uniformity across the cross section of the oxide. The CTE mismatch between the coating constituents and the oxide should be such that it minimizes the stress in the system. The oxide should have excellent creep related properties, good fatigue strength, resistance to impact failure.

The prominent mode of degradation in a TBC is the oxidation of bond coat leading to the formation of thermally grown oxide (TGO) resulting in generation of stress due to coefficient of thermal expansion mismatch. The pegging of the TGO of the bond coat induces stress resulting in cracking of the top coat [141]. The formation of NiO and $Ni(Cr,Al)_2O_4$ as a part of the TGO serves as the weakest link between the top coat and the bond coat [142]. The failure is commonly initiated at the TGO and is instigated by thermal expansion mismatch [143]. A TGO typically consists of α-Al_2O_3, Cr_2O_3 and few Ni based oxides and spinels, the TGO film is durable when the oxide is fine grained and uniformly thick and a smooth transition from mixed oxide zone to uniform oxide zone. A TGO with a thickness of less than 2 μm are observed to have high stability and effectively resists substrate oxidation [144]. The TGO thickness greater than 5 μm introduces stresses of very high magnitude and results in premature failure of the coating system by coating detachment [145]. The TGO thickness increases with increase in exposure duration, the growth is high during the initial 200 h which tends to gradually slow down [77]. The growth of TGO is also affected by cyclic thermal exposure, the surface roughness of the TGO was found to increase with increase in number of thermal cycles [146]. The oxidation kinetics and degradation study of electron beam physical vapor deposition also emphasizes on the increase in bond coat- TGO interface thickness leading to failure of the coating [147]. The presence of yttrium and hafnium increases the adhesion of the bond coat with the surface [9, 148]. The absence of hafnium in the bond coat accelerates the formation of delaminating crack resulting in spallation of the coating. The doping of Hf also results in the formation of uniform TGO, the presence of 0.5 at% Hf in NiAl bond coat improved the cyclic oxidation behavior of the coating. Increasing the Hf to 1.5 at% resulted in formation of straight TGO as compared to the lower Hf at% doping and undoped coating [149].

The degradation of coating in presence of molten deposits in molten salt environment is called hot corrosion. Hot corrosion occurs when the metals, alloys, ceramics are covered with molten salt in a temperature range of 700–925 °C [150, 151]. Sodium sulphate salts are deposited on the surface of the component as a result of the reaction between the sodium salts and sulfur present in the flue gases. Hot corrosion which occurs at the melting point of the salt deposits is called type 1 hot corrosion. The hot corrosion occurring at a temperature lower than the melting point of salt deposits is type 2 hot corrosion. The molten salt destroys the protective oxides and renders the coating ineffective due to internal oxidation and sulfidation. The degradation occurs starts with the breakdown of the protective oxide followed by propagation stage during which the molten salt gains access to the naked metal enhancing the rate of degradation. Figure 8

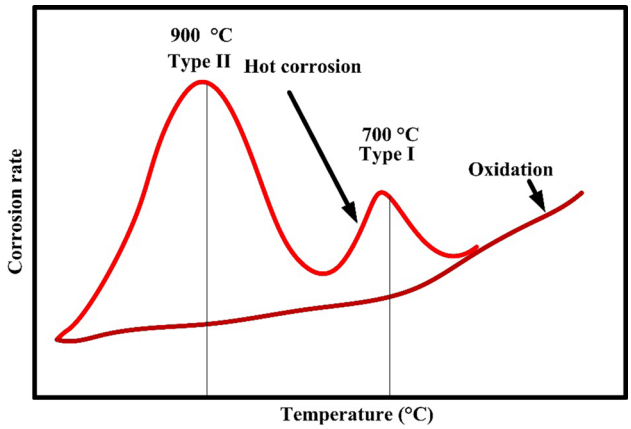

Fig. 8 Schematic representation of type 1 and type 2 hot corrosion rate

shows the hot corrosion rate of type 1 and type 2 superimposed on oxidation rate variation with temperature.

Hot corrosion has a pronounced effect on the life of TBC, the degradation enhances manifold due to the presence of molten deposits. The deposits affect the bond coat along with the top this occurs due to the inherent porosity of the APS yttria stabilized zirconia (YSZ). The molten salt penetrates the YSZ through the cracks, pores and reacts with the bond coat. The metallic nature of the bond coat makes it highly susceptible to degradation under the influence of molten salt [1, 152–154]. The phase transformation of YSZ from tetragonal to monoclinic results in degradation of the TBC. This phase transformation is accompanied by a 4% volumetric change which increases the stress in the top coat. In some cases, accelerated sintering of YSZ also occurs which increases the Youngs modulus resulting in degradation of life of the coating [155]. The infiltration salt deposits introduce stress due to coefficient of thermal expansion mismatch between the molten deposit and the YSZ. The in-service failure of YSZ can be classified into four regimes of the time vs temperature plot. The four regimes in the order of increasing time and temperature are infant mortality, erosion, molten salt infiltration and thermochemical phenomena. The thermochemical phenomenon includes phase transformation, bond coat degradation, and sintering [156].

4.5 Life Prediction Model for TBC

A study at National Bureau of Standards (NBS) conducted in 1942 can be consideredto the first application of ceramic coatings. The work was aimed to protect the ordinary steel (ex. SAE 1020) used in exhaust stacks. A thin ceramic coating was recommended by the NBS to increase the life of this component that operated inred-hot condition and was subjected to quick heating and cooling. The coated steel met the drastic requirements without suffering any significant damage. The successes of this development lead to an increased interest in ceramic coatings for heat-resistant alloys.

The success of this ceramic coating work on the development of coatings for turbine blades and similar applications began at the NBS with financial assistance from National Advisory Committee for Aeronautics (present-dayNASA). The work started by developing coatings using silica, feldspar, alumina, titania, and other refractory ceramics, the composition was designed to minimize the coefficient of thermal expansion (CTE) mismatch between the substrate and coating. The feedstock was melted in gas-fired pot furnace and the coating was applied by spraying. Alumina, chromium oxide, and cobalt oxide singly and in combination successfully met the CTE requirement and were then subjected to various tests simulating the actual operating conditions [157]. This coating provided a high degree of protection to the alloys at a higher temperature and strong corrosive environment. The coatings increased the surface oxidation resistance of the substrate and were found to enhance the life of the component. The frit coating was tested on turbine blades for next few years [158]. However, the problem related to spalling of coating persisted and was highlighted in XLR99 rocket engine application.

This coating failure caused the overheating of the substrate which resulted in boiling of liquid ammonia in circulation for the regenerative purpose. The ammonia vapors being highly corrosive made the failure of the conduit inevitable. The scope for improvement in the then used Rokide Z coating was marginal and hence emphasis was laid on development of a new coating system. A graded coating was proposed and developed with 100% metal composition near the substrate to 100% ceramic at the surface. The metal to ceramic ratio along the thickness was maintained by feeding appropriate premixed feedstock in the plasma arc during the coating deposition process. The graded coating consisted of a molybdenum primer with Nichrome graded to zirconia. The results of the engine tests were encouraging, the coating showed a significant improvement for at least 5 min as compared to the RokideZ coating. The graded coating improved the adherence of thermal barrier coating (TBC) and prevented premature failure of the combustion chamber. Drummond and Alves [152] studied different plasma sprayed and slurry-based TBC. The coatings consisted of thin layers of metal and metal oxides. The metal oxides included alumina, zirconia, chroma and molybdenum, nichrome, tungsten were metals included in the graded coating. Tungsten rich coating system exhibited superior performance based on coating loss. The surface coating having alumina-nichrome graded coating with zirconia top coat showed the minimum depth of surface cracking and coating loss. The coatings were tested in hydrogen plasma and the majority of the tested coatings did not flake or spall under the severe testing conditions.

Drummond and Alves [152] further investigated the structural integrity and thermal conductivity of three different metallic-ceramic composite coating. The thermal conductivity of these coatings made them effective for thermal barrier coating applications. The performance of the coating which was not sprayed at optimum spray parameters did not exhibit satisfactory performance. The graded zirconia and molybdenum layered coating with zirconia and hafnia outer layer functioned for 17 cycles and exhibited less than 1 percent damage after 213 s of testing. The type 3 and 5 coating, a graded series of alumina, nichrome, molybdenum and zirconia, nichrome, molybdenum respectively functioned without failure for 6 cycles and 182 cycles of operation. The lower thermal conductivity of plasma sprayed coating was a result of higher contact resistance between the plasma sprayed particles.

Hodge et al. [159] studied the effect of stabilizers on the performance of the zirconia coating. The performance of CaO, MgO, and Y_2O_3 stabilized ZrO_2 in the molten salt environment was evaluated. A few Specimens were prepared by slip casting and few by air plasma spray technique. Metal/ceramic graded coating was deposited by plasma spraying on button shaped specimens. These button-shaped specimens were subjected to burner rig isothermal and cyclic tests at 926 °C. Stabilized zirconia in the form of slip cast tubes was immersed in molten Na_2SO_4 at 898 °C for a period of 100, 200, 300, and 1000 h. Some of the slip cast hollow specimens were coated with graded metal/ceramic system and subjected to burner rig test. The coated slip cast hollow specimens were subjected to a temperature gradient by maintaining 1040 °C and 815 °C on the external and internal surfaces respectively. The slip cast specimens did not exhibit gross salt penetration and suffered minor deterioration during the 1000 h of exposure. In case of plasma sprayed samples, the deterioration of the metal phase the underlying alloy occurred only for those specimens that did not contain vapor deposited CoCrAlY initial coating. The samples subjected to cyclic test suffered greater damage than the isothermally treated specimens. This increased attack is a result of coating spallation caused by the thermal shock that rendered the coating unprotective. The coating in addition to being corrosion resistant should remain intact during thermal cycling or must contain an adherent innermost protective layer. The coated hollow specimen exhibited greatest temperature gradient for ceramic outer layer. The ceramic layer showed presence of cracks as compared to the specimen with metallic outermost layer. The metal or ceramic phase of the coating showed little sulfidation attack even when the external layer were at very high temperatures. The TBC system deposited by plasma spray technique offered high corrosion resistance and improved thermal insulation. However efforts should be taken to improve the coating adherence either by varying the composition or modifying the microstructure.

Stecura [160] developed a bilayered plasma sprayed TBC system for protecting air-cooled gas turbine components. The most promising TBC system had a bond coat Ni–16Cr–6Al–0.6Y (wt%) and ZrO_2–$12Y_2O_3$ (wt%) as the top coat. This TBC system withstood 3200 and 275 cycles without cracking or spallation. ZrO_2 stabilized with 3.4%MgO and 5.4%CaO was also subjected to cyclic heating and cooling, the MgO and CaO stabilized coating cracked and spalled after 460 and255 cycles respectively at 975 °C. The failures occurred in the top coat at bond coat-top coat interface. At Mach 0.3 burner rig test the substrate temperature was found to be 140–190 °C lower than that of the ceramic surface exposed to the flame. The ZrO_2–$12Y_2O_3$ TBC system deposited on air-cooled turbine blades withstood 3200 and 275 cycles at 1280 °C and 1490 °C. This coating system was most resistant to erosion and cracking with very high adherence as compared to that of other coating systems. The TBC ceramic layer became thinner with time in the burner rig test but no gross spallation was observed.

Drummond and Alves [152] in 1979 studied the effect of compositional changes on the performance of TBC system. TBC system included a bond coat containing 16Cr, 6Al and 0.15 to 1.08Y % with zirconia top coat. The top-coat composition was varied by introducing 4.0 to 24.4 wt% Y_2O_3. The coatings were evaluated in furnace cyclic heating and cooling from 990 to 280 °C and 1095 to 280 °C. The solid specimens, air-cooled gas turbine blades were tested with the surface exposed to 1580 °C temperature. The test was carried out using a natural gas-oxygen torch test rig, in case of turbine blades, a flame operating at Mach 1.0 was employed. The results indicated best thermal barrier coating had a bond coat combination of Ni–16.4Cr–5.1Al–0.15Y and Ni–17.0Cr–5.4Al–0.35Y. The top coat with 6.2% and 7.9% Y_2O_3 provided best TBC performance. These coating withstood for 1500 and 100 h of cyclic life at 990 °C- 280 °C and 1095 °C- 280 °C respectively in furnace thermal cyclic test. In natural gas-oxygen torch test rig system the solid specimen withstood 590 to 790 cycles at 1200–100 °C. In case of Mach 1.0 burner rig test the two systems withstood over 1400 and 2000 cycles without failure at 1480 °C and 1470 °C of surface temperature respectively.

The failed coatings exhibited formation of cracks that grew parallel to the bond-top coat interface. The solid specimens exhibited cracks that were at 90 °C to the bond coat and the coating on the turbine blades were either at 90 °C or at 35–45 °C angle to the bond coating. The application of TBC was extended to engine parts to determine the potential benefits of coating system. The application was extended to turbine vanes, high pressure advanced core engine parts. The coating system was tested at high pressure and high temperature inlet condition to simulate the gas inlet condition. The durability test results of the ceramic coating on the

turbine blades and vanes in furnace and during turbine operation were encouraging. The coating successfully reduced the metal temperature by 390 °C at a constant gas coolant flow rate. The ceramic coating thickness can be increased to decrease the coolant flow rate without affected the underlying substrate temperature. Zirconia thickness of 195 µm increased the allowable turbine inlet gas temperature by around 500 °C. The coating thickness in the range of 51 µm to 510 µm showed adequate adherence to the vane and blade metal walls without any sign of spallation for 40 cycles of heating and air cooling [161].

The plasma sprayed zirconia coatings performance was studied after post plasma spray treatment. Zaplatynsky [72] evaluated the performance of laser treated ZrO_2 coating in cyclic oxidation and corrosion burner rig tests. ZrO_2 coating with two thickness 200 µm and 400 µm were laser treated to partially melt the coating surface. The cyclic life of the coating was unaffected by the laser treatment. However the laser treated samples showed fourfold life improvement in corrosion test carried out in flame containing 100 PPM of sodium fuel equivalent. The life of the samples was inversely related to the thickness of the coating. The laser glazed zirconia top coat showed striations, segmentation and mud flat cracking produced by melting. The laser-induced segmentation did not have any effect on the performance of the coating. The crack propagation in top coat, near and parallel to the top coat-bond coat interface lead to failure of the TBC system.

The failure of the coating is a critical factor in deciding the component life. The need for a model to predict the life of a coating was addressed by miller et al. A oxidation based approach was used for life prediction and a model was developed to related the heating cycle duration and cycles to failure [152]. The model was used to realte the number of cycles to failure to the duration of heating cycle. The model was based on the assumption that oxidation is the predominant time dependent factor that limits the life of the coating. The oxidation induced strain promotes the crack growth in the ceramic coating. The radial component of the CTE mismatch strain (ε_r) is considered as the strain that directly contributes to the crack porogation at the interface. The critical amount of oxidation was characterised by a critical specific weight gain (w_c) such that tahe coating fails in the first cycle itself. The value of w_c will depend on many factors like the coating thickness, porosity, properties of the bond coat, and specimen geometry. The boundary conditions can be expressed as:

Cycle to failure (N_f) = 1, when w = w_c and Mismatch strain (ε) = Critical strain (ε_r)
Cycle to failure (N_f) = Maximum number of cycles to failure (N_{fo}), when w = 0and $\varepsilon = \varepsilon_r$
Cycle to failure (N_f) = 1, when w = 0and Mismatch strain (ε) = failure strain (ε_f). The failure strain is a material specific property at a particular location for a particular specimen. The differecne between the effective strain (ε_e) and CTE msimatach strain (ε_r) will be directly proprtional to the speicifc weight gain due ot growht of oxide. The final expression for coating life was given as

$$\sum_{N=1}^{Nf}\left[\left(1-\frac{\varepsilon r}{\varepsilon f}\right)\left(\frac{wn}{wc}\right)^{m}+\frac{\varepsilon r}{\varepsilon f}\right]^{b}=1 \tag{1}$$

The term b represents the subcriticla crack growth exponent which have been reported as 17 for plasma sprayed 8% YSZ coating. The papameter m refers the order of relationship beween the strain and the wieht gain due ot oxidation.

Thermomechanical life prediction models have been developed to predict life of aiur plasma sprayed and electron beam deposited TBC. Fracture mechanics and statstical methods have been developed to relate the tensile and the compressive strain for the thermalmechcnial life model. The life prediction models were developed for four modes of failure zirconia spallation, bond coat oxidation, molten salt film damage, and carbon particle errosion [152]. Life predication models by Drummond and Alves [152] have also been developed to evaluate the effect of coating edges, oxidation of bond coat, creep behaviour of bond coat, top coat and bond coat thicknee. The model predicts that the TBC life can be imporoved by minimizing the edge effect of the coatings. The bond coat oxidation producing strain due to volumetric increment, it also reduces the strain due to thermal expansion mismatch between the bond coat and the top coat. A model was developed using experiment analysis and finite element analysis to relate normal strain and the shear strain develoed during thermal cycles. The model is as given below:

$$\Delta\varepsilon_{RZ}+0.4\Delta\varepsilon_{R}=0.084N_{f}^{-1.445} \tag{2}$$

$\Delta\varepsilon_{RZ}$ and $\Delta\varepsilon_R$ represents the shear strain range and the normal strain range and N_f is the number of cycles to failure.

The spallation of the coating is dependent both on the cyclic thermal loading and the thermal exposure. A damage accumulation model was developed by Drummond and Alves [152] using both tha cyclic and exposure induced damage. The cyclc dammage was related to the inelastic strain and the time dependent exposure damage was related to the oxidation kinetic of the bond coat. The cyclic plastic strain ($\Delta\varepsilon_p$) is related to the single cycle failure strain $\left(\Delta\varepsilon_t\right)$ by manson–coffin type relation, as given below:

$$\textit{Number of inelastic strain cycles to failure}(N)=K\left(\frac{\Delta\varepsilon_t}{\Delta\varepsilon_p}\right)^{b} \tag{3}$$

A spallation life model for EBPVD ceramic coating was developed by Meier et al. [162] by analysing the

quantitative cyclic thermal spallation life data, interfacial oxide characteristics, coating adherence and other mechanical properties of the coating. The spallationwas found to occur by cracking at TGO at the top coat-bond coat interface. The properties of the coating were measured by tensile and compression testing of thin walled single crystal superalloy tubes deposited on the external diameter. The elastic strength was found to be around 310–345 MPa at 537 °C. A bond coat oxidation model was developed for the EB PVD coating by analysing the furnce test results. The following oxide scale thickness model was developed:

$$\delta = \left\{ \exp\left[Q\left(\frac{1}{T_O} - \frac{1}{T} \right) \right] t \right\}^{n} \tag{4}$$

The terms δis the oxide thickness in microns, Q is the ration of apparent activation energy and universal gas constant, T_O is the temperature constant = 2423.7 K and t = time (s).

The TBC life was modelled by developing a correlation between the spallation life and the TGO growth characteristics and the TGO strain levels during a burner rig test. The burner rig teat cycles were simulated to meet strain emphasied, oxide growth emphasised and mixed mode condition. The strain emphasied condition was simulated by rapid thermal cycling and the oxide growth was simulated by conducting the test at maximum temperature of the cycle. The inelastic deformation is a critical life parameters for ceramic coatings however the burner rig test for EBPVD coating indicate that the coating reamins elactic. The cracking was found to occur at the TGO and bond coat interface hence the modelling efforts have used elastic strain for development of life predication model. The mechanical strain in the TGO was calcuated by substracting the strain due to free eleongation of the TGO from the total strain.

$$\varepsilon(T) = \varepsilon_{SS}(T) - (\delta L(T)/L) \tag{5}$$

ε = is the mechanical strain in the TGO, ε_{SS} = total strain of the substrate

The cyclic life is represented by power law as below,

$$Cyclic\ life(N) = A\Delta\varepsilon^{b} \tag{6}$$

A is the empirical normalising constant and b the the power law coefficient. The predictive capability of the life prediction model was verified using three verification tests.

A Fracture mechanics based approach was used by Traeger et al. [163] to develop an finite element model based on the crack growth predication. The FE model was influenced by the thickness of the TGO (d_{TGO}), bond coat-top coat interface surface roughness. The behaviour of the material was assumed to be fully elastic and the thermo mechanical stresses are assumed to be generated as a result of coefficent of thermal expansion mismatch. The stress distribution was expressed by the analytical expression in the form given below;

$$g(t) = \left[R\frac{\alpha}{L} e^{\left(-\delta \frac{A}{d_{TGO}(t)}\right)} + 1.803\left(\frac{A}{L}\right)^{0.5} \times \left[\left(\frac{L}{\xi A}\right)^{0.5} \frac{d_{TGO}(t)}{A} - 1 \right] \right] \frac{\sigma_O}{\sigma_{av}} e^{-\left(\frac{\beta R}{L}\right)} Y \frac{A}{L^{1.5}} \tag{7}$$

The terms A, L and R represent the roughness amplititute, wavelength and the radial coordinate of the TGO. The term σ_{avg} represents the thermal stress induced due to CTE mismatch between the coating and the substrate. The development of anlyatic model for stress is follwed by modelling of crack growth. The initial cracks are assumed to be formed under the influence of tensile stress. The state of stress is altered when a certain amount of oxide growth occurs at the BC-TC interface. The cracks are generated at peak tips when the state of stress become compressive, this crakcs grow and lead to spallation of the coating. The crack growth rate is represnted by a power law as given below;

$$\frac{da}{dt} = \nu_O \sigma^m a^{\frac{m}{2}} \left(\frac{Y}{K_{Ic}} \right)^m \tag{8}$$

ν_O is the growth rate when $K = K_{IC}$ and is assumed to be 7.6×10^{-5} m/s, m = 18 and the critical stress inentsity factor(K_{IC}) is 1 MPa $\sqrt{m}$. Y is the geometry function, σ is the stress rangel for a cycle $\sigma_{max} - \sigma_{min} = \sigma$. The TGO thickness d_{ox} is expressed as below;

$$d_{ox}(t,\ T) = \left(ko\ e \frac{-E_{TGO}}{K_B T} t \right)^{0.5} \tag{9}$$

E_{TGO} is the activation energy and k_o is the pre-factor, K_B is the boltzman constant and T is the interface temperature. The youngs modulus and the stress in the coating varies with the number of thermal cycles. The increase in stress after each cycle is assumed to be the reason for failure of the coating. The expression representing the youngs modulus is as given below;

$$E(t,T) = E_O + A_O e^{\frac{-E_{\sin t}}{k_B T}} t^{0.5} \tag{10}$$

E_{sint} is the activation energy and A_o is the pre-factor.

The stress in the coating at the interface is a summation of the thermal stress, CTE mismatch stress and stress due to addition external strains.

$$\text{Thermal stress } \left(\sigma_{therm}\right) = \Delta\alpha\Delta T\left(E(t,T) - E_O\right) g(t) \tag{11}$$

$$\sigma_{sub} = \frac{17MPa \times 0.005m}{r\sigma_{av}} E(t,T)\Delta\alpha\Delta T \quad (12)$$

$$\sigma_{\max} = \left[R\frac{\alpha}{l}e^{-\delta}\sigma_o e^{\left[\frac{-\beta R}{L}\right]}Y\frac{A}{L^{1.5}}\right] \times -\frac{\eta\varepsilon_mech}{\Delta\alpha_{sub} - TGO\Delta T} \quad (13)$$

The FEM model developed with stress distribution patterns as shown above was found to have reasonable aggrement with the experimental results.

A fracture mechancis based life prediction model was developed by Ni et al. [164] the failure location was found to be TBC–TGO interface and occurred as a result of crack propogationa and coalescence. During thermal cycling a large amount of stress is developed at the top coat and bond coat interface. This stress inititates crack at the metal ceramic interface which propogates and results in failure of the coating. The TGO thickness is time dependent related as shown below;

$$\text{Thickness of TGO } \left(d_{TGO}\right) = k_p t^m \quad (14)$$

"m" is the time exponent and "k_p"is the temperature dependent growth rate. The critical crack length is 8.8 mm and is given by the expression given below:

$$C \approx 2.21\, D_{TBC}\left(\frac{E_{TBC}}{\left(1 - V_{TBC}^2\right)}\right)^{0.5} \quad (15)$$

E_{TBC}, V_{TBC} is the youngs modulus and the poissions ratio.

$$\sigma = \frac{E_{TBC}}{1 - V_{TBC}}\left(\alpha_{TBC} - \alpha_{sub}\right)\Delta T$$

α_{TBC} and $\alpha_{sub}$is the coefficient of thermal expansion of TBC and the substrate.

The crack propogation rate was related to the stress intensity factor (ΔK) with power n

$$\frac{dc}{dN} = A^*(\Delta K)^n \quad (16)$$

$$\Delta K = \Delta\sigma\sqrt{\pi C} \quad (17)$$

$\Delta\sigma$ is the stress range at minimun temperature, C is the length of crack. A life oprediction model is developed by fitting the experimental data which is represented by the log form of the paris law with an calculated values of A^* and n.

The bond strength of the TBC is affected by the thermal fatigue experienced by the coating. Song et al. [165] developed a life prediction model based on the variation of bond strength of the coating with the number of cycles. The Eqs. (18), (19) and (20) shows the relationship between the bond strength for sample after N thermal fatigue cycles with 15 min, 20 min, and 30 min of heating duration in a cycle. The effect of duration of heating was also considered, the model is given as below in Eq. (23)

$$\sigma_{15\min} = -0.215 \times N + 62.5 \quad (18)$$

$$\sigma_{20\min} = -0.271 \times N + 62.5 \quad (19)$$

$$\sigma_{30\min} = -0.355 \times N + 62.5 \quad (20)$$

$$a = a_o \times f(t) = -0.215\,(1 + 0.0442t) \quad (21)$$

$$\sigma = -0.215(1 + 0.0442t) \times N + 62.5 \quad (22)$$

$$At\ failure\,(\sigma = 0)\,N = \frac{-\sigma_O}{a_o \times f(t)} = \frac{290.70}{1 + 0.0442t} \quad (23)$$

Wei et al. [166] developed a numerical life prediction model by predicting the failure stress of the top coating. The numerical results were found to be consistant with the experimental results. The TGO thickness was related to the dwell peroid (1 h) and the thermal cycles using the followign relation.

$$TGO\ thickness\,(h) = 0.86 \times (no.\ of\ thermal\ cycles)^{0.3} \quad (24)$$

The coating was assumed to fail when an delaminating crack was visible at the surface of the coating. The FEM model was based on the following failure reasons (1) failure due to roughnening of TC/TGO/BC interface (2) excessive TGO growth (3) creep (4) sintering due to high temperature exposure (5) CTE mismatch between the TC, TGO and the BC. The life prediction model development methodology was developed in 3 steps, the first step involved the analysis of SEM images to obtain a cumulative probablity of microstructural characteristics. The FEM model was build on the boundary conditions obtained from TGO thickness. The next step was to consider the creep, oxidation, and sintering and calculate the stress and strain at hold temperature (Tmax) and on cooling refered as $\sigma_{(Tmax)}$, $\varepsilon_{(Tmax)}$ and $\sigma_{(Tmin)}$, $\varepsilon_{(Tmin)}$ respectively. The third step was to finalize the failure criterion of the coating which was (peak stress) $\sigma_{(Tmin)}$ > failure stress. The lifetime of the TBC lies in between $\frac{1}{t_{hold}}\left(\frac{\Delta h(i-1)}{B}\right)^{\frac{1}{m}}$ and $\frac{1}{t_{hold}}\left(\frac{\Delta h(i)}{B}\right)^{\frac{1}{m}}$ for the TGO thickness when varied from Δh(i-1) to Δh(i).

5 Future Work

The hot corrosion resistance of YSZ thermal barrier coating is lower, no attempts are seen in the literature to develop a composite coating to improve the hot corrosion behaviour

of thermal barrier coating. Effect of whisker reinforcement on the dispersion of particles on fracture toughness of the TBC has not been reported in the literature. There are very few stress relaxation techniques for determining the residual stress in the TBC. A cost effective and convient method for quantifying the residual stress in the coating. There is very less literature availble in life prediction model in presence of molten salt environment.

6 Conclusions

The future of thermal barrier coating (TBCs) require design and development of advanced materials for protecting the turbines and other engine components resulting in improvement in better fuel efficiency, engine reliability and reduced downtime of the unit. The study of various aspects like the residual stress, high temperature performance of TBCs is very critical in development of newer coating material and TBC systems. The development of life prediction model of coatings is essential before the newly developed coatings are used in actual working environment. The understanding of failure mechanism and process induced stresses is very important before application of next generation advanced TBCs.

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